

Field Experiments Technical Report

Author: Abdullah Maruf, Inframind Labs

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Subject: Field-data evidence, limitations, and controlled-collection readiness for the GSSI 51600S ground-penetrating radar archive.

Executive Summary

The field-data work produced a useful but carefully bounded result. The current GSSI 51600S ground-penetrating radar archive can support measured two-dimensional quality evidence: profile import, reflector cue screening, relative timing checks, waveform-shape consistency, and broad-window signal contrast.

The same archive does not yet support field full waveform inversion, three-dimensional field modeling, calibrated cover-depth recovery, target radius recovery, or amplitude-calibrated claims. The missing pieces are physical measurement controls: absolute timing references, known target geometry, known target depth and size, surveyed profile geometry, amplitude references, accepted metadata, and accepted real file checksums.

The current field archive is therefore useful as a measured two-dimensional quality benchmark. It is not yet an inversion data set.

Key Findings

Question	Finding	Meaning
Is the archive usable?	Yes, for measured two-dimensional quality evidence.	The four radar profiles can support timing, waveform-shape, and signal-contrast checks.
Is the archive ready for full waveform inversion?	No.	The data do not include absolute timing, target truth, amplitude calibration, or surveyed geometry.
Is the archive a three-dimensional field survey?	No.	The available files behave as independent line profiles, not as a surveyed three-dimensional grid.
What is the next valid field milestone?	A controlled collection with real metadata and reference files.	The collection workflow is ready, but measured files and accepted metadata are still missing.

What Data Are Available?

The available field archive contains four radar profile files collected with a GSSI 51600S ground-penetrating radar system. Each profile is a two-dimensional radar line profile. The current archive does not include the survey information needed to treat those files as a three-dimensional grid.

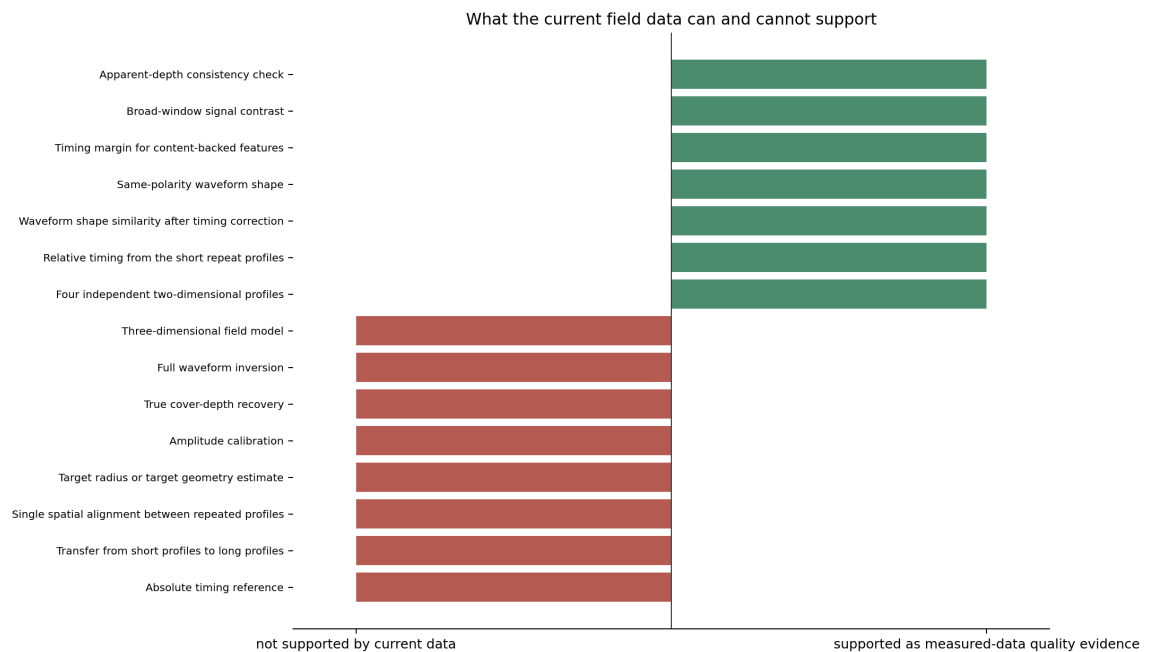
The data contain visible radar responses that can be screened and compared. The initial screening found nineteen raw reflector cues. After traceability checks, eleven of those cues became support anchors. A support anchor is a reflector cue with enough consistency and context to support a limited technical claim.

Data Item	Count	What It Means
Radar profile files	4	Four readable measured line profiles are available.
Raw reflector cues	19	Nineteen visible radar responses were identified during screening.
Traceable support anchors	11	Eleven cues were consistent enough to support limited quality evidence.
Short-profile timing anchors	3	Three short-profile anchors were inside the supported timing envelope.
Content-backed short-profile anchors	2	Two short-profile anchors had enough waveform content to support the strongest timing and waveform-shape claims.
Long-profile pattern anchors	8	Eight long-profile anchors showed pattern consistency, but not timing transfer from the short profiles.
Must-have controls for inversion currently satisfied	0 of 5	The archive is not yet defensible for calibrated inversion.

Evidence Boundary

The field work separates what the data support from what they do not support. This boundary is the most important result. The current archive supports measured quality evidence, not calibrated inversion.

Figure 1. Evidence boundary for the current field data. Green items are supported as measured-data quality evidence. Red items remain unsupported by the current archive and require new controlled measurements before promotion.



What the Current Data Supports

Supported by Current Data	Not Supported by Current Data
Four readable GSSI 51600S radar profile files	Three-dimensional subsurface model
Nineteen raw reflector cues	Absolute time-zero calibration
Eleven traceable support anchors	True cover-depth recovery
Short-profile relative timing: 0.127701 nanoseconds	Target radius or target geometry recovery
Content-backed timing residual: 0.009823 nanoseconds	Amplitude-calibrated interpretation
Waveform-shape correlation at accepted settings: 0.939 minimum	Full waveform inversion
Broad-window signal contrast: 5.05 times minimum root-mean-square ratio	High-performance computing escalation for field inversion

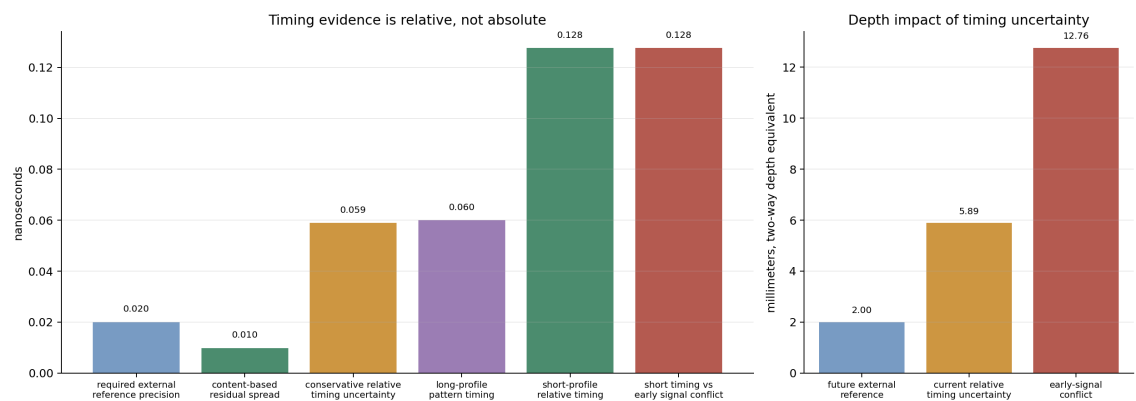
Timing Evidence

The most important positive result is the short repeat-profile timing result. The short repeat profiles support a relative timing offset of 0.127701 nanoseconds. The content-backed residual spread is much smaller, about 0.009823 nanoseconds, which is strong enough for a measured-data quality check.

This does not create an absolute timing calibration. It means the short repeat profiles can be compared after a relative alignment. A separate external timing reference is still required before depth inversion or absolute time-zero claims.

Figure 2. Timing evidence in nanoseconds and depth-equivalent units.

The future external timing reference requirement is tighter than the current relative timing uncertainty. The early/direct signal conflicts with the content-backed short-profile timing, so it cannot be used as a substitute for absolute time zero.

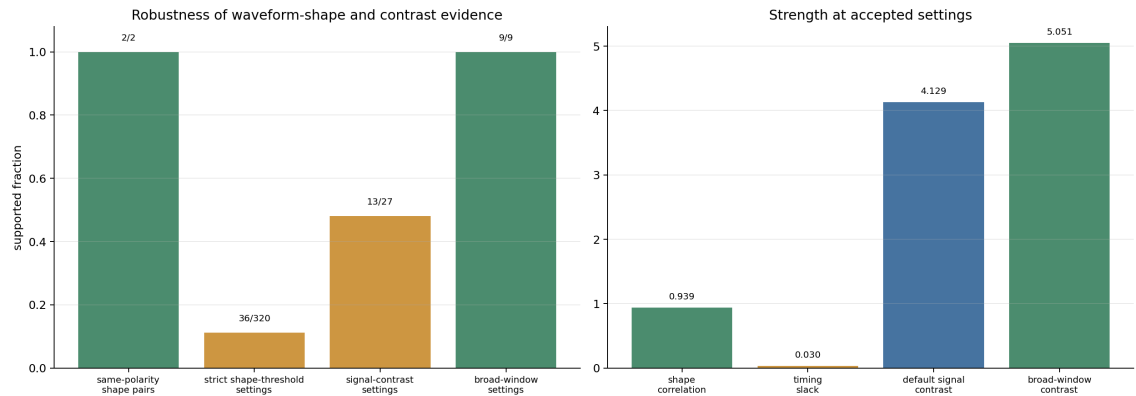


Waveform Shape and Signal Contrast

The short-profile waveform-shape evidence is strong at accepted settings. The minimum corrected waveform correlation is 0.939. The same-polarity waveform shape check is supported for both content-backed short-profile pairs.

The result is still a quality check, not a calibrated physical inversion. A strict threshold sweep supported both pairs in only 36 of 320 tested settings. Signal contrast was also not fully window-invariant: all windows were supported in 13 of 27 tested combinations. The broad-window contrast case was stronger, with all 9 broad-window combinations supported and a minimum root-mean-square contrast ratio of about 5.05.

Figure 3. Waveform-shape and contrast robustness. The left panel shows which fraction of tested settings supported the result. The right panel shows the strength of the accepted settings. The data support a waveform-shape and broad-window contrast quality claim, not amplitude calibration.

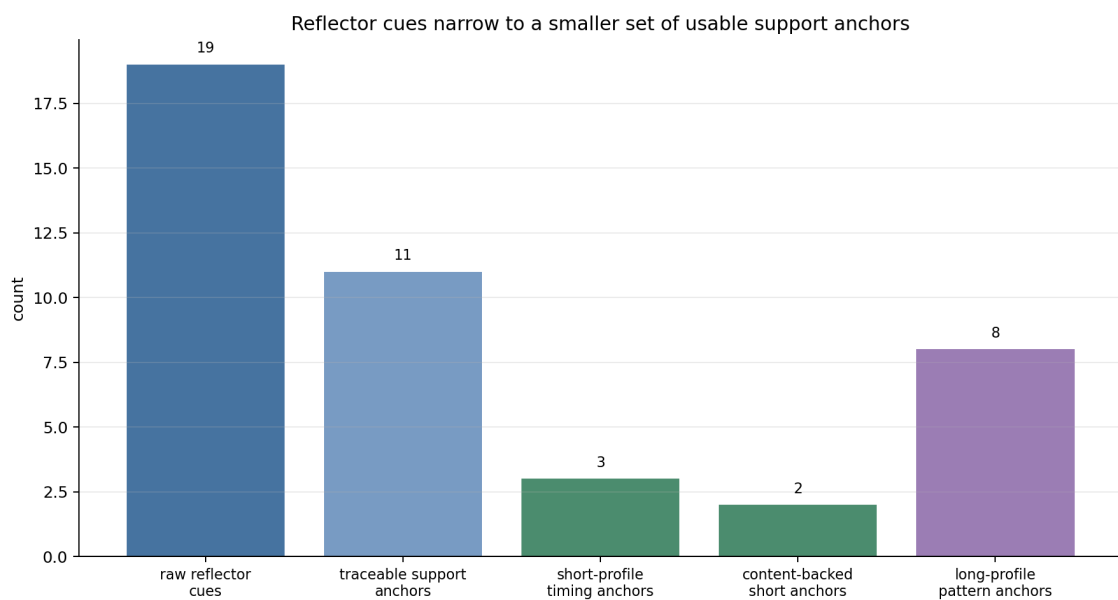


Reflector Cue Traceability

The reflector-cue analysis separates raw visual cues from claim-bearing anchors. Nineteen raw cues were found, but only eleven became traceable support anchors. Three short-profile timing anchors were inside the supported timing envelope, and two of those were content-backed.

The long profiles provided eight pattern anchors, but those anchors did not support transferring the short-profile timing correction to the long profiles.

Figure 4. Reflector cue funnel. Raw reflector cues narrow to a smaller set of support anchors. This prevents overclaiming from visual reflector patterns alone.



Why Field Inversion Is Not Yet Defensible

Full waveform inversion requires more than a visually plausible radar response. The current archive is missing the physical controls needed to connect measured waveforms to a calibrated subsurface model.

The current archive satisfies 0 of the 5 must-have controls needed for defensible inversion:

- absolute time-zero reference;
- surveyed profile and target geometry;
- known target radius or diameter;
- known cover depth with dielectric or velocity calibration;
- reference amplitude calibration.

Without these controls, a full waveform inversion result could look precise while still being physically unanchored.

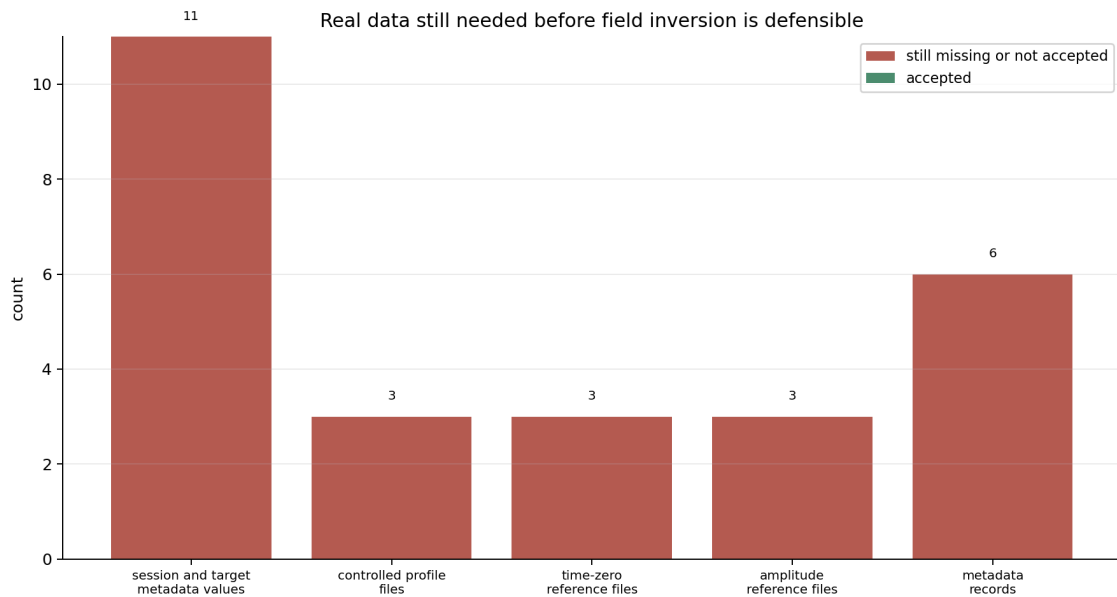
Controlled Collection Readiness

The later field work converted the blockers into a practical controlled collection package. The package defines the measured files, metadata, archive layout, checksum ledger, intake checks, and acceptance checks needed for a future controlled field collection.

The planned collection still needs real data. The current accepted count is zero for every required real-data group.

Required Item	Needed	Accepted Now	Why It Matters
Session, operator, target, and profile metadata	11	0	These values identify how, when, and where the radar data were collected.
Controlled profile files	3	0	Repeated controlled profiles are required to check repeatability.
Time-zero reference files	3	0	External timing references are required for absolute timing and depth calibration.
Amplitude reference files	3	0	Amplitude references are required before amplitude-calibrated comparisons or full waveform inversion.
Metadata files or records	6	0	These records provide file-level traceability and make later review repeatable.

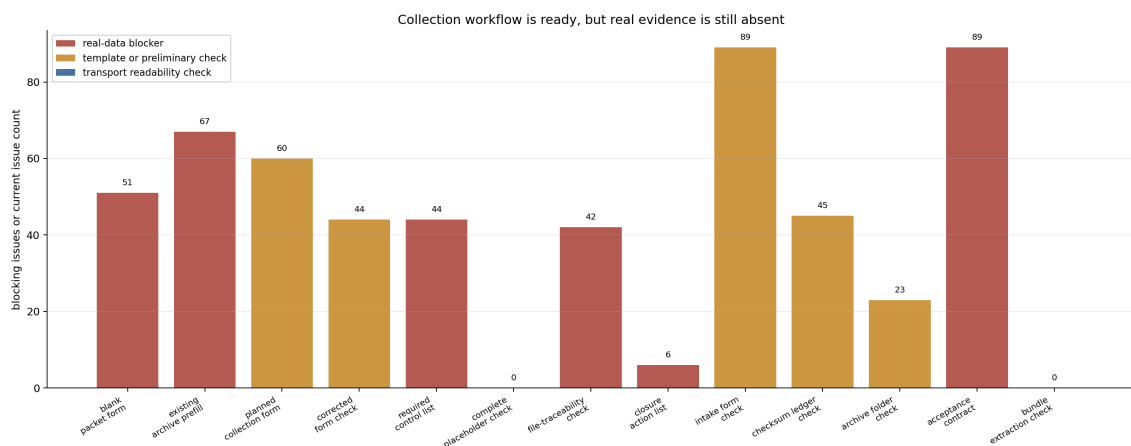
Figure 5. Real data gap before field inversion. The current workflow requires eleven metadata values, nine real radar files, and six metadata records before the field archive can be accepted for scientific escalation.



Collection Acceptance Path

The acceptance pathway was tested with templates and placeholder data. Those tests show that the workflow can pass when all required fields are present, but they do not create measured evidence. A file-traceability check still blocks placeholder values, missing files, and unresolved file references.

Figure 6. Collection acceptance pathway. Several forms and preliminary checks are operationally ready, but the real-data blockers remain. The final transport readability check confirms that the handoff bundle can be unpacked, not that real measured field evidence has been accepted.



Progression of the Field Work

Analysis Stage	Question Answered	Main Result	Supported Use	Limit
Import and cue screening	Could the radar files be read and screened?	Four radar profile files were readable, and nineteen raw reflector cues were found.	Measured radar-profile quality screening.	No confirmed target labels, depth recovery, or inversion.
Survey scope and repeatability	Was the archive a three-dimensional survey?	The archive behaved as independent two-dimensional line profiles. The short repeat profiles were the strongest timing pair.	Two-dimensional timing and repeatability checks.	No surveyed three-dimensional layout, target radius, or cover-depth recovery.
Short-profile relative timing	Could the short repeat profiles support a timing correction?	A relative timing offset of 0.127701 nanoseconds was supported for the short repeat profiles.	Relative timing alignment for measured-data quality checks.	No absolute timing calibration and no transfer to long profiles.
Long-profile pattern evidence	Did the short-profile timing transfer to the long profiles?	The long profiles had a repeatable pattern shift, but it did not match the short-profile timing correction.	Long-profile pattern context.	No short-to-long timing transfer and no inversion-ready timing model.
Timing-family separation	Did all timing cues represent the same physical timing reference?	The early/direct signal behaved like a common system response, while content-backed short-profile timing remained a separate relative signal.	A clear timing boundary for reporting.	No absolute time-zero calibration and no calibrated depth claim.
Waveform shape and signal contrast	Did waveform shape and signal contrast make the archive inversion-ready?	Waveform-shape and broad-window contrast evidence were strong enough for a quality claim, but not for calibrated inversion.	Measured waveform-shape and contrast quality evidence.	No amplitude calibration, target geometry, target radius, true cover depth, or full waveform inversion.
Controlled collection design	Which real controls are missing?	The existing archive satisfied zero of five must-have controls needed for defensible inversion.	Design of a future controlled field collection.	The existing archive remains unsuitable for inversion or high-performance computing escalation.

Analysis Stage	Question Answered	Main Result	Supported Use	Limit
Complete-form and file-traceability checks	Could complete forms pass without real files?	Complete placeholder forms can pass format checks, but file-traceability checks still block placeholder values and missing files.	Acceptance checklist for a future controlled collection.	Placeholder values cannot be promoted to measured field evidence.
Collection-day handoff package	Is the future collection workflow operationally ready?	The handoff package is transport-readable and checksummed, but real archive acceptance is still false.	Operator handoff for future controlled data collection.	No scientific field claim or inversion until real files and metadata pass all checks.

Conclusions

The current field archive is a valuable measured-data quality benchmark. It supports careful statements about two-dimensional radar profile quality, relative timing, waveform-shape consistency, and broad-window signal contrast.

The current field archive is not yet an inversion data set. It lacks absolute timing, target truth, surveyed geometry, amplitude calibration, and accepted real file traceability. These missing controls block field full waveform inversion, three-dimensional field modeling, calibrated cover-depth recovery, and target-radius recovery.

The next scientifically valid field milestone is a controlled data collection with real measured files and accepted metadata. Once that controlled collection passes archive folder checks, checksum verification, intake review, format validation, and file-traceability validation, the field data can be reconsidered for calibrated inversion work.

Terms and Abbreviations

Term	Plain-English Meaning
Ground-penetrating radar	A radar method that sends electromagnetic pulses into a material and records reflections from internal features.
GSSI 51600S	The radar system used to collect the field data reviewed here.
DZT file	The raw radar profile file format produced by the GSSI system.
Time zero	The timing reference used to decide when the radar pulse effectively starts in the recorded trace.

Term	Plain-English Meaning
Amplitude calibration	A measurement that relates recorded signal strength to a known reference, so amplitude comparisons can be physically interpreted.
Full waveform inversion	A computational method that estimates subsurface properties by matching the full measured waveform, not just picked reflection points.
High-performance computing	Large-scale computing resources used for expensive simulations or inversions.
Checksum	A file fingerprint, such as a SHA-256 value, used to confirm that a file has not changed.
Metadata	Descriptive information about the measurement, such as operator, time, target details, profile geometry, and file identity.
Reflector cue	A visible radar response that may correspond to an internal object or material boundary.
Support anchor	A reflector cue that has enough traceability and consistency to support a limited technical claim.